

```
docker --version
```

```
sudo apt update
```

```
sudo apt install apt-transport-https ca-certificates curl software-properties-common -y
```

```
sudo systemctl start docker
```

```
sudo systemctl enable docker
```

```
sudo docker run hello-world
```

```
mkdir my-python-app
```

```
cd my-python-app
```

```
nano app.py
```

```
nano Dockerfile
```

```
nano requirements.txt
```

```
docker build -t myapp-image .
```

```
docker run -d -p 5000:5000 --name myapp myapp-image
```

```
ls
```

```
docker login -u ankith05
```

```
docker tag myapp-image ankith05/my-python-app:latest
```

```
docker push ankith05/my-python-app:latest
```

```
git config --global user.name "Ankith0505"
```

```
git config --global user.email "ankithshetty0505@gmail.com"
```

```
git config --list
```

```
git clone https://github.com/Ankith0505/jenkins-docker.git
```

```
cd jenkins-docker
```

```
docker build -t my-python-app .
```

```
docker images
```

#Dockerfile

FROM python:3.11-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY app.py .

EXPOSE 5000

CMD ["python", "app.py"]

#Jenkinsfile

pipeline {

agent any

environment {

DOCKER_IMAGE = "ankith05/my-python-app1"

DOCKER_TAG = "latest"

}

stages {

stage('Build Docker Image') {

steps {

sh 'docker build -t \$DOCKER_IMAGE:\$DOCKER_TAG .'

}

}

stage('Login to DockerHub') {

steps {

withCredentials([usernamePassword(
credentialsId: 'dockerhub-credentials',
usernameVariable: 'USER',
passwordVariable: 'PASS'

])) {

sh 'echo \$PASS | docker login -u \$USER --password-stdin'

}

```

    }
}
stage('Push Image') {
    steps {
        sh 'docker push $DOCKER_IMAGE:$DOCKER_TAG'
    }
}
stage('Deploy Container') {
    steps {
        sh '''
            docker stop myapp-container || true
            docker rm myapp-container || true
            docker run -d -p 5001:5000 --name myapp-container $DOCKER_IMAGE:$DOCKER_TAG
        '''
    }
}
}
}
}

```

```

# app.py
from flask import Flask
app = Flask(__name__)
@app.route("/")
def home():
    return "Hello! Docker is running this Python app successfully "
if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)

```

```

#requirements.txt

```

```

flask

```

```
sudo apt update
```

```
sudo apt install maven -y
```

```
mvn archetype:generate -DgroupId=com.example.app \
```

```
-DartifactId=demo-app \
```

```
-DarchetypeArtifactId=maven-archetype-quickstart \
```

```
-DinteractiveMode=false
```

```
cd demo-app
```

```
ls
```

```
nano pom.xml
```

```
nano src/main/java/com/example/app/App.java
```

```
nano src/test/java/com/example/app/AppTest.java
```

```
mvn clean install
```

```
mvn exec:java -Dexec.mainClass="com.example.app.App"
```

```
git init
```

```
git add .
```

```
git commit -m "Initial commit"
```

```
git branch -M main
```

```
git remote add origin https://github.com/<your-username>/demo-app.git
```

```
git push -u origin main
```

```
sudo rm jenkins.war
```

```
sudo rm -rf ~/.jenkins
```

```
wget https://get.jenkins.io/war-stable/latest/jenkins.war
```

```
java -jar jenkins.war --httpPort=8090
```

#App.java

```
package com.example.app;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.apache.commons.lang3.StringUtils;

public class App {

    private static final Logger logger = LoggerFactory.getLogger(App.class);

    public static void main(String[] args) {

        String name = "CI/CD Pipeline";

        if (StringUtils.isNotBlank(name)) {

            String message = greet(name);

            logger.info(message);

            System.out.println(message);

        } else {

            logger.error("Name is empty!");

        }

    }

    public static String greet(String name) {

        return "Hello, " + name + "! Welcome to Maven CI/CD Demo.";

    }

}
```

#AppTest.java

```
package com.example.app;

import org.junit.Test;
import static org.junit.Assert.assertEquals;

public class AppTest {

    @Test

    public void testGreet() {

        String result = App.greet("Student");

        assertEquals("Hello, Student! Welcome to Maven CI/CD Demo.", result);

    }

}
```

```
#Jenkinsfile

pipeline {
    agent any

    tools {
        maven 'Maven3'
        jdk 'JDK21'
    }

    stages {
        stage('Checkout') {
            steps {
                git branch:'main',
                    url:'https://github.com/Ankith0505/Maven-project.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean compile'
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'
            }
        }

        stage('Package') {
            steps {
                sh 'mvn package'
            }
        }

        stage('Run Application') {
            steps {
```

```

        sh 'mvn exec:java -Dexec.mainClass="com.example.app.App"'
    }
}
}
post {
    success {
        emailtext (
            subject: "SUCCESS: ${JOB_NAME} #${BUILD_NUMBER}",
            body: "Build succeeded!\nCheck: ${BUILD_URL}",
            to: "ankithshetty0505@gmail.com"
        )
    }
    failure {
        emailtext (
            subject: "FAILED: ${JOB_NAME} #${BUILD_NUMBER}",
            body: "Build failed!\nCheck: ${BUILD_URL}",
            to: "ankithshetty0505@gmail.com"
        )
    }
}
}

```

#pom.xml

```

<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-
v4_0_0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.example.app</groupId>
    <artifactId>demo-app</artifactId>
    <packaging>jar</packaging>

```

```
<version>1.0-SNAPSHOT</version>
<name>demo-app</name>
<url>http://maven.apache.org</url>
<dependencies>
  <dependency>
    <groupId>ch.qos.logback</groupId>
    <artifactId>logback-classic</artifactId>
    <version>1.4.14</version>
  </dependency>

  <!-- JUnit for testing -->
  <dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>4.13.2</version>
    <scope>test</scope>
  </dependency>

  <!-- Logging -->
  <dependency>
    <groupId>org.slf4j</groupId>
    <artifactId>slf4j-api</artifactId>
    <version>2.0.9</version>
  </dependency>

  <!-- Apache Commons -->
  <dependency>
    <groupId>org.apache.commons</groupId>
    <artifactId>commons-lang3</artifactId>
    <version>3.14.0</version>
  </dependency>
```


</dependencies>

<build>

<plugins>

<!-- Compiler Plugin -->

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-compiler-plugin</artifactId>

<version>3.11.0</version>

<configuration>

<source>17</source>

<target>17</target>

</configuration>

</plugin>

<!-- Test Plugin -->

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-surefire-plugin</artifactId>

<version>3.1.2</version>

</plugin>

</plugins>

</build>

</project>